

TimeTensors Experiment 0: Reference Orders of Error

1 Goal

Before comparing training choices, a compact reference establishes the scale of univariate forecasting error under the exact dataset splits, normalization, and evaluation stride used by TimeTensors.

Experimental goal. Compare a non-trainable baseline, a task-trained neural model, and a pretrained foundation model without turning the study into an architecture leaderboard.

2 Protocol

Persistence repeats the final look-back value. PatchTST uses instance normalization, normalized-MSE training, random sampling, batch size 256, and drops users with accessible constant look-backs. Chronos-2 is frozen and uses its local checkpoint. All methods share the same chronological split and an evaluation stride equal to the forecast horizon.

The default study profile uses six datasets, (168, 24), (504, 168), and (504, 504), and seeds 1–3. The launcher is `04_reference.slurm`. The launcher builds data-space MSE and worst-user-decile MSE tables; per-user artifacts remain available for follow-up analyses. Deterministic methods may have zero seed variance.